A Hybrid Vision Transformer and Feature-Based Model for Mitotic Figure Classification in Glioma

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***Abstract*—Accurate classification of mitotic figures in glioma histopathology images is critical for reliable tumor grading and informed treatment planning. In this study, we introduce HybridNet, a hybrid deep learning framework that combines the high-level semantic capabilities of a pre-trained Vision Transformer (ViT) with the domain-specific discriminative strength of handcrafted features. Specifically, four handcrafted feature sets, local binary patterns, texture descriptors, morphological attributes, and color statistics, are extracted to complement the ViT's learned representations. These engineered features are concatenated with the partially fine-tuned ViT features to enhance classification performance and model interpretability. An ablation study was conducted to evaluate the contribution of each handcrafted feature group, revealing consistent performance gains with each module addition. In addition to HybridNet, we benchmark several conventional machine learning classifiers trained solely on handcrafted features, as well as standalone deep learning architectures including ResNet50, DenseNet121, EfficientNet-B0, and ViT, to establish robust performance baselines. The proposed method is evaluated on a curated dataset of glioma cell crops using five-fold cross-validation, achieving a mean classification accuracy of 91.75% with consistent performance across folds. The results underscore the complementary value of integrating handcrafted and deep features and highlight HybridNet's potential as an interpretable and computationally efficient solution for mitotic figure classification in digital pathology workflows.**

***Index Terms*—HybridNet, Handcrafted Features, ViT, Image Classification, Glioma, Mitotic figures**

I. Introduction

Gliomas are the most prevalent and aggressive form of primary brain tumors, accounting for approximately 81% of malignant brain neoplasms in adults and nearly 30% of all brain tumors worldwide [1]. Accurate subclassification of gliomas is critical for determining prognosis and guiding personalized therapeutic strategies. However, such stratification remains challenging due to the tumors' highly infiltrative nature and the difficulty of distinguishing mitotic from non-mitotic cells in routine histopathological evaluation [1]. Mitotic figures, which indicate active cellular proliferation, occur infrequently and are often visually indistinguishable from non-mitotic nuclei, particularly in the presence of staining variability, imaging noise, and subjective interpretation by pathologists. These challenges introduce inconsistencies in manual assessments and have motivated the adoption of machine learning (ML) and deep learning (DL) approaches to automate mitosis detection. Nonetheless, conventional DL models often struggle to reliably capture the subtle morphological and textural cues essential for accurate classification, limiting their effectiveness in this high-stakes clinical task.

To address these limitations, prior studies have employed a range of advanced feature extraction and data augmentation techniques. Texture-based descriptors such as Gray-Level Co-occurrence Matrix (GLCM) and Gray-Level Run-Length Matrix (GLRLM) have been used to characterize tumor heterogeneity [2]–[4], while handcrafted statistical features, including edge sharpness, intensity histograms, and standard deviation, have been explored for their complementary diagnostic value [2]. Methods such as Local Binary Patterns (LBP) and Gabor filters have further enhanced textural analysis, often in combination with dimensionality reduction techniques like Principal Component Analysis (PCA) [3], [5]. In parallel, spectroscopic methods such as Fourier Transform Infrared (FTIR) analysis have been investigated to provide biochemical insights into glioma tissue [6]. The diversity of data sources and modalities underscores the breadth of current approaches. While many studies utilize whole-slide images (WSIs) from The Cancer Genome Atlas (TCGA) [2], [7], [8], others integrate MRI and FLAIR modalities for multi-scale or multi-modal analyses [3]. Deep learning–oriented research typically leverages large-scale patch-based datasets with aggressive data augmentation strategies to promote generalizability [5], [7], [9]. For instance, the MIDOG 2022 dataset has served as a benchmark for mitotic cell detection, although limitations remain in terms of external validation and reproducibility [8], [10]. Conversely, multi-institutional datasets employed in [4], [9] have shown promise in enhancing model robustness.

Data augmentation remains a cornerstone for addressing class imbalance and dataset scarcity. Common techniques—such as flipping, cropping, rotation, and translation—help simulate biological variability, while stain normalization methods (e.g., Macenko normalization [7]) and artifact removal procedures are essential for reducing confounding factors. Synthetic data generation has also been explored to enrich underrepresented classes, particularly for rare tumor subtypes [3]. Among the DL architectures explored, convolutional neural networks (CNNs) such as ResNet,

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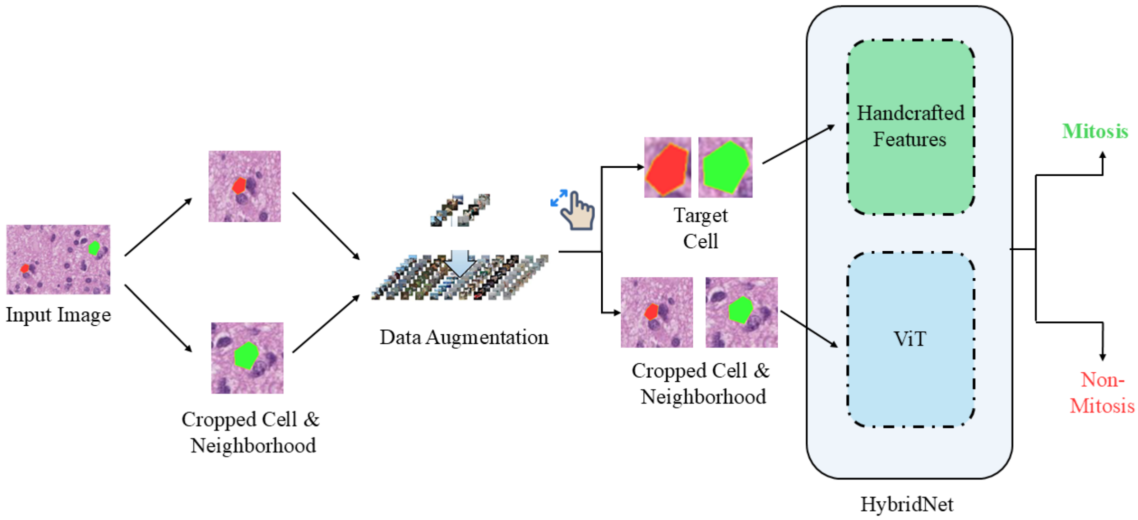

Fig. 1. Overview of the data flow and processing pipeline for mitotic figure classification.

DenseNet, and VGG-19 [4], [5], [7], [9] remain dominant, often initialized with ImageNet weights and subsequently fine-tuned on glioma-specific datasets. Classical ML classifiers, including support vector machines (SVMs), random forests, and decision trees, have also been utilized, frequently in conjunction with recursive feature elimination (RFE) or LASSO-based selection for interpretable modeling [3], [4]. Hybrid approaches that integrate DL features with handcrafted descriptors have shown compelling gains in classification performance. For instance, [5] achieved an accuracy of 98.25% using VGG-19 with hybrid features, while [6] reported 99% using an artificial neural network on FTIR spectra. The recent Glioma-MDC 2025 Challenge [11] highlighted the state-of-the-art in mitotic figure detection. The winning submission, SEMA [12], achieved perfect F1-scores using a semi-supervised cross-magnification attention mechanism and pseudo-labeling. Other top-performing models utilized optimized ConvNeXt V2 pipelines (F1 $>$ 0.98) [13], Vision Transformers with stain normalization (F1 = 0.9824) [14], and CNN ensembles augmented with super-resolution techniques (F1 = 99.15%) [15]. While these advances reflect cutting-edge design and engineering, a critical shortcoming persists: most models fail to fully leverage the complementary strengths of handcrafted features and deep neural representations.

This study addresses this gap by proposing a novel hybrid architecture, HybridNet, which fuses high-level semantic features from a pre-trained Vision Transformer (ViT) with a curated set of handcrafted features (GLCM, LBP, Gabor, and statistical descriptors). Our goal is to enhance classification accuracy, model robustness, and interpretability-key prerequisites for clinical deployment in computational pathology. The remainder of this paper is organized as follows: Section II details our methodology, including dataset preparation, feature extraction, and modeling strategies across ML, DL, and hybrid paradigms. Section III presents a comprehensive evaluation of the proposed and baseline models. Finally, Section IV discusses the implications of our findings and outlines potential directions for future research.

II. Methodology

In this section, we describe the process of dataset curation, feature extraction, and the evaluation framework for both traditional ML and DL models. Based on the insights gained from these baseline experiments, we introduce our proposed HybridNet architecture for mitotic figure classification, the overall pipeline of which is illustrated in Fig. 1.

A. Dataset Preparation and Feature Extraction

The Kaggle Glioma-MDC 2025 dataset [11] contains high-resolution WSIs with mitotic and non-mitotic regions annotated. Image preprocessing began with the removal of improperly labeled polygons to ensure annotation consistency. The dataset was then split into mitosis and non-mitosis classes. To address class imbalance prior to classical machine learning model training, random under-sampling was performed by downsampling the majority class to match the size of the minority class, thus ensuring a balanced dataset. This was necessary to prevent classifier bias and maintain equal class representation during training. For feature extraction, each cell-centered patch was processed to derive morphological, color, and texture descriptors, including GLCM and LBP features. Additionally, Macenko Stain Normalization was applied to reduce variability due to histological staining differences. Finally, all extracted features were normalized using Min-Max Normalization to scale them within a uniform range and enhance model performance.

TABLE I
AVERAGE PERFORMANCE METRICS OF MACHINE LEARNING MODELS. BOLD VALUES INDICATE THE BEST-PERFORMING RESULTS FOR EACH METRIC.

Model	Acc. (%)	Prec. (%)	Recall (%)	F1-Score (%)
SVM	61.2	62.0	59.1	60.5
SVM-RFE	60.3	61.4	55.9	58.5
Decision Tree	58.7	58.5	59.8	59.1
Extreme Boosting	**64.7**	**65.1**	**66.4**	**65.7**
Random Forest	62.5	63.0	63.4	63.2
Linear Model	61.0	61.3	60.1	60.7

B. Machine Learning Models for Mitosis Detection

Upon feature extraction, six conventional ML classifiers were evaluated with 5-fold cross-validation: Support Vector Machine (SVM), SVM Recursive Feature Elimination (SVM-RFE), Decision Tree (DT), Extreme Gradient Boosting (XGBoost), Random Forest (RF), and Logistic Regression. XGBoost was the best, with 64.7% accuracy, 65.1% precision, and 66.4% recall, with outstanding mitosis detection at low false positives. Random Forest was good (62.5% accuracy) but lacking in false negatives and positives, while SVM was more accurate (62.0%) but poorer in recall (59.1%), and thus better for identifying non-mitosis cells. SVM-RFE and Decision Tree were poor, especially recall, with the lowest accuracy of 58.7% being shown by Decision Tree. Logistic Regression worked moderately with accuracy of 61.0%. These results are summarized in Table I.

C. Deep Learning Models for Mitosis Detection

In addition to machine learning models, deep learning architectures were explored, taking advantage of both local and global feature extraction. The images were resized so that the target cell occupied 50% of the image without harming the cell and its background. Various data augmentation techniques such as, rotation, flipping, brightness/contrast, and elastic deformation were applied to the training set to prevent class imbalance and improve generalization. However, no data augmentation was applied in the test and validation sets to ensure a balanced evaluation. Deep learning models used were ResNet50, DenseNet121, EfficientNet-B0, and Vision Transformer (ViT). The models were pre-trained on ImageNet, while CNN-based models (ResNet50, DenseNet121, and EfficientNet-B0) were trained from scratch, but the ViT was fine-tuned. In order to conserve computational resources, 2-fold cross-validation was chosen instead of 5-fold cross-validation. This approach allowed for a quicker evaluation of model performance, enabling the selection of the best model to build upon for further experimentation.

Among the deep learning models, the performance of ViT was optimal at 89.2% accuracy, 89.3% precision, 89.1% recall, and 89.2% F1-score. Such high performance indicates that the global context information provided by ViT significantly enhanced mitosis detection. Similarly, the performance of ResNet50 was also good with 87.6% accuracy and balanced performances in precision, recall, and F1-score. EfficientNet-B0 equally did a great job with 87.2% accuracy, and DenseNet121 had the lowest performance among the deep learning models with 86.8%. Table II shows the summary of these results. These results confirm that ViT, with its ability to capture global context, provided superior performance compared to the CNN models. However, CNNs like ResNet50 and EfficientNet-B0 also demonstrated strong performance, indicating the importance of both local and global context in mitosis detection.

TABLE II
AVERAGE PERFORMANCE METRICS FOR DEEP LEARNING MODELS.

Model	Acc. (%)	Prec. (%)	Recall (%)	F1-Score (%)
EfficientNet	87.2	87.3	87.1	87.1
ResNet	87.6	87.7	87.6	87.6
DenseNet	86.8	86.9	86.8	86.8
ViT	**89.2**	**89.3**	**89.1**	**89.2**

D. Hybrid Model: ViT Fusion with Handcrafted Features

Despite the impressive results achieved by ViT, we hypothesized that the model might still overlook crucial local information specific to the target cell itself. To address this, we proposed HybridNet, a hybrid model that combines the power of ViT with handcrafted feature extraction modules. The goal was to expxlore deep learning's ability to learn global contextual features from large datasets while also incorporating handcrafted features that capture subtle, domain-specific characteristics of the target cell. This fusion of global and local feature learning allowed HybridNet to better distinguish between mitotic and non-mitotic cells, making it well-suited for complex histopathological image analysis tasks. HybridNet structure includes employing a pre-trained (fine-tuned) ViT model (*vit_base_patch16_224*) as the primary global feature extractor. Four hand-designed feature extraction modules are used along with the ViT model for extracting significant local features. These include the LBPModule, which extracts local texture patterns using Local Binary Patterns; the TextureModule, which identifies overall texture variations through convolutional layers; the MorphologyModule, which focuses on morphological attributes that assist in distinguishing mitotic from non-mitotic cells; and the ColorStatsModule, which computes color statistics to achieve staining variability typical of histological images.

To further customize the model to the target cell, we carried out a zoom-in preprocessing. This pre-processing step eliminates the background of the image, capturing only the region of interest surrounding the target cell, then zooms it to the original image size to simulate a zoomed-in appearance. This is done so that the model can focus on local detailed features, thus the most important characteristics of the target cell are better preserved. Once the deep features from ViT and handcrafted features from the four modules are obtained, they are combined in a single feature vector. This unified representation is input into fully connected layers, which include training stability by batch normalization and a dropout rate of 0.6 to prevent overfitting. The final layer is for binary classification to distinguish between mitosis and non-

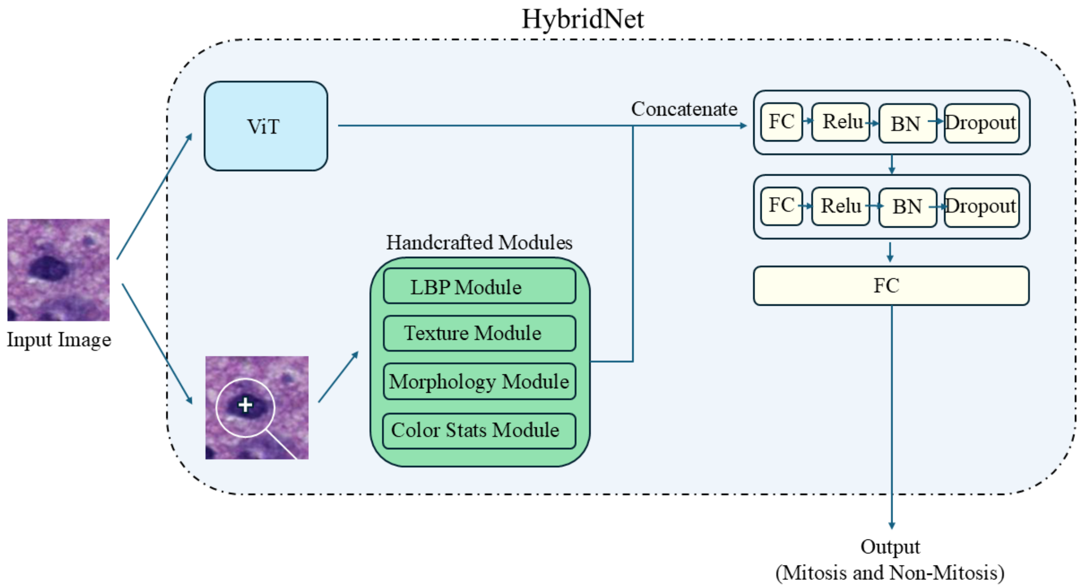

Fig. 2. Schematic overview of the HybridNet architecture for mitotic figure classification.

mitosis. Figure 2 described the HybridNet architecture flow for classifying Mitosis/Non-Mitosis.

To further understand the contribution of each handcrafted module within HybridNet, we conducted an ablation study. In this study, we progressively removed or added individual modules (LBP, Texture, Morphology, and ColorStats) from the HybridNet architecture to quantify their individual impact on performance. The results of this analysis are presented and discussed in Section III.

III. Results and Discussion

The HybridNet model achieved outstanding results on the test dataset with 91.75% accuracy, 91.18% precision, 92.5% recall, and 91.6% F1-score. Figure 3 shows training vs validtion accuracy for HybridNet Model using the 5 fold. All these scores are higher than the solo ViT model performance, representing the success of the hybrid approach in the detection of mitosis. The confusion matrix shown in Fig. 4 also reveals that HybridNet was highly successful in the classification of mitotic and non-mitotic cells, showing its high capacity to classify the two classes accurately with minimal misclassification.

To evaluate the contribution of each handcrafted module in the HybridNet architecture, we conducted an ablation study. Table III shows that we started from the baseline ViT-only model, which achieved 87.2% accuracy, then incrementally added handcrafted modules,starting with Local Binary Patterns (LBP), followed by Texture, Morphological, and ColorStats features. The full HybridNet model, which includes all four modules, reached 91.75% accuracy. This progression highlights the complementary value of handcrafted features to

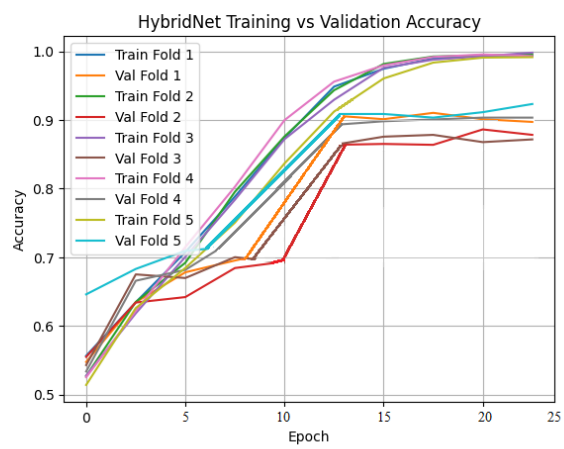

Fig. 3. Training and validation accuracy curves of the HybridNet model over training epochs.

ViT, especially the significant boosts seen after integrating texture and morphology information. The consistent increase in accuracy across these configurations further emphasizes the robustness of each feature set. Figure 5 presents a clear comparison of accuracy progression across the ablation configurations.

The ablation results further demonstrate that each handcrafted feature group contributed to performance improvement, validating the hypothesis that local image characteristics supplement the ViT's global context learning. Among all modules, the TextureModule and MorphologyModule showed

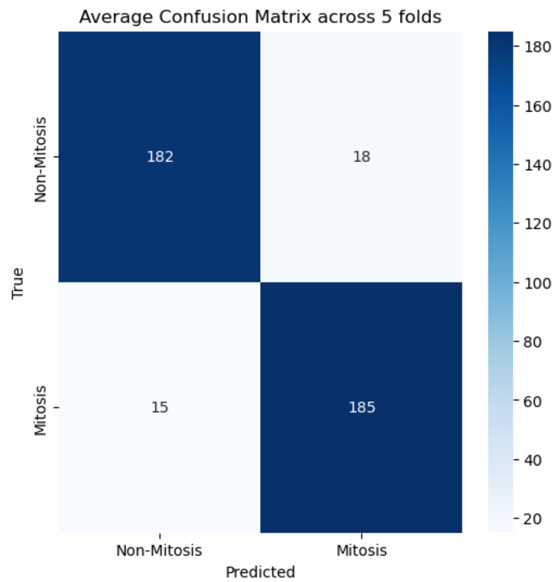

Fig. 4. Confusion Matrix for HybridNet Model.

TABLE III
ABLATION STUDY RESULTS HIGHLIGHTING PERFORMANCE DIFFERENCES.

Model	Acc. (%)	Prec. (%)	Recall (%)	F1-Score (%)
ViT Only	89.2	89.3	89.1	89.2
ViT + LBP	89.9	90.2	89.7	89.9
ViT + Morphology	90.4	90.6	90.3	90.4
ViT + ColorStats	90.0	90.1	89.9	90.0
ViT + TextureModule	90.7	90.8	90.5	90.6
ViT + All (HybridNet)	**91.75**	**91.18**	**92.5**	**91.6**

the highest individual impact, likely due to their sensitivity to subtle structural variations critical for mitosis recognition. LBP and ColorStats also provided meaningful performance gains, although to a lesser extent.

The complete HybridNet, combining all handcrafted descriptors, achieved the highest overall performance, confirming the synergy between handcrafted and deep features. These findings reinforce our claim that hybrid designs, when carefully engineered, can outperform purely deep learning–based pipelines by integrating domain-specific knowledge. This is particularly important in clinical pathology, where interpretability and targeted feature relevance play a critical role.

Although HybridNet's performance was not as good as the F1-scores of top submissions in the Glioma-MDC 2025 (MIDOG) challenge, such as the SEMA method [12] or models like ConvNeXt V2 and Swin Transformer [13]–[15], it is significantly better in simplicity, interpretability, and flexibility. These merits make HybridNet a highly practical and novel solution to the problem of mitosis detection. Unlike the competition models that strictly adhere to deep learning-based feature extraction, HybridNet integrates handcrafted knowledge and learned features of deep learning models, endowing interpretability and greater flexibility to make future changes.

IV. CONCLUSION

The present work presents an end-to-end method for mitosis classification in histopathological images, contrasting classical machine learning models and deep learning models. While classical models like XGBoost and Random Forest performed well when local features were used, the ViTmodel based on global context outperformed them. To boost performance even more, we introduced HybridNet, a novel fusion of ViT and handcrafted feature modules that learn domain-specific features like texture, morphology, and color statistics. The hybrid approach achieved an accuracy of 91.75%, outperforming all previous approaches. The results highlight the importance of combining both local and global features in improving diagnostic accuracy. The ablation study confirmed that each handcrafted feature module contributed distinct improvements, underscoring the complementary strengths of domain-specific descriptors and transformer-based representations. Future work will focus on improving image quality using encoder-decoder networks like U-Net or autoencoders to recover lost fine details, especially in cropped or low-quality images. Additionally, dynamic cropping techniques using object detection algorithms like YOLO or Faster R-CNN will help localize target cells, improving the model's robustness to off-center or partially blocked cells. This will enables an integrated end-to-end pipeline for both detection and classification tasks.

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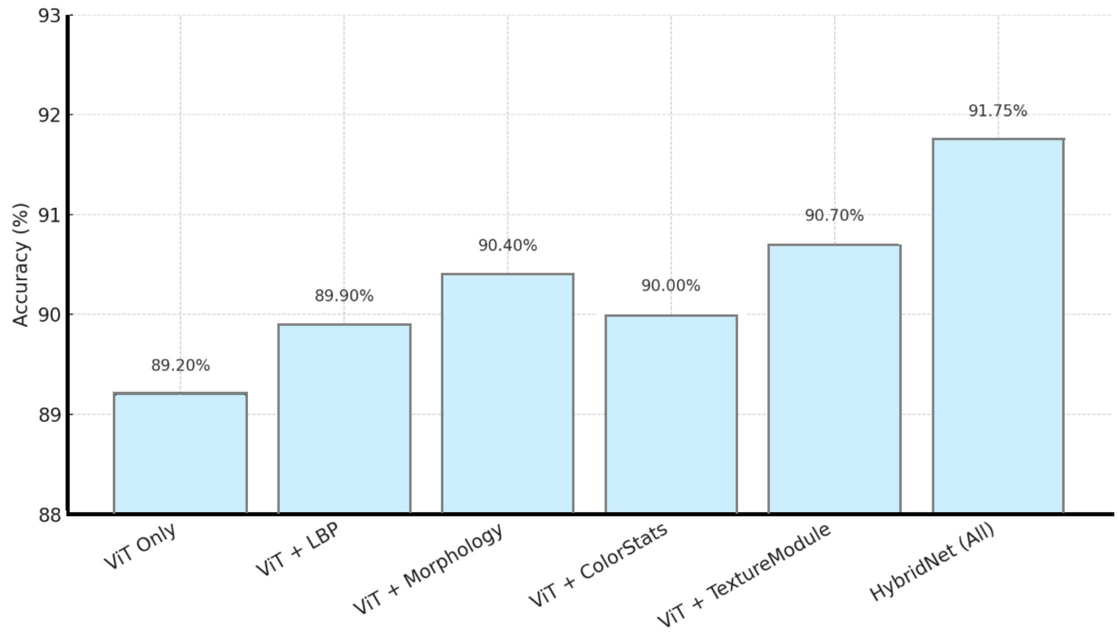

Fig. 5. Accuracy progression from ViT-only to full HybridNet.

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